

RYAN CONROY

Golden, CO

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Education

Colorado School of Mines

Aug. 2025 – May 2029

Bachelor of Science in Electrical Engineering – GPA: 4.00 / 4.00

Golden, CO

Experience

Colorado School of Mines Formula SAE

Aug. 2025 – Present

Electrical Engineer – PCB Design

Golden, CO

- Own PCB design end-to-end across both the EV and IC Formula SAE cars, taking boards from schematic to fabricated, assembled, and bench-verified hardware installed on the vehicle.
- Coordinate with other subsystems on PCB enclosure, connector placement, and harness routing so board outlines, mounting, and clearances are finalized before layout.
- Verify designs against FSAE electrical rules and IPC-2221 clearance and creepage limits, revised through team design reviews in a shared Altium 365 workspace.

Ace Hardware

Apr. 2022 – Present

Sales Associate

Anthem, AZ

- 4+ years. Trusted and dependable representative supporting back-end operations and customer facing responsibilities.

Projects

High-Voltage Presence Indicator | Altium, LT8316, Isolated Flyback

2026

- Designed a safety-critical indicator in a 450V battery that trips when the tractive bus exceeds 60V, with no software in the control path per FSAE rules.
- Built an LT8316 no-opto isolated flyback (504V peak to a 12V secondary) with an 8:1:1 transformer, 900V MOSFET, and TVS drain clamp; sized the EN/UVLO divider from 0.1% thin-film resistors so the hysteresis band lands inside the required 60V-on / 50V-off window.
- Designed the isolation boundary so a single-point fault on the high-voltage side can't propagate to the low-voltage domain, per FSAE rules.

Telemetry Ground Station | Altium, ESP32-S3, Raspberry Pi, ESP-NOW

2026

- Designed an ESP32-S3-WROOM-1 board that receives lap-timing data from trackside gates over ESP-NOW at ~200m, forwards it to a Raspberry Pi 5 over UART, and logs to microSD over SPI, giving the team lap data with no reliance on trackside networking infrastructure.
- Implemented the power tree: TPS56339 buck (5V to 3.3V), TPS2116 priority mux so the board runs from Raspberry Pi or bench USB-C, PTC fuses, and USBLC6-2SC6 ESD protection at the connector, with the outline matched to the Raspberry Pi form factor.

Current & Voltage Sensor | Altium, Hall-Effect Sensing, Analog Design

2026

- Designed a bus-powered board measuring $\pm 150\text{A}$ bidirectional current with an ACS758 Hall-effect sensor and 0–15V bus voltage through an MCP6001-buffered divider, with TVS and resettable-fuse input protection.
- Ran the dissipation analysis that justified replacing a switching regulator with a single LM1117 linear regulator ($\sim 160\text{mW}$), cutting part count, board area, and switching noise near the analog front end.

FPV Quadcopter & Custom Flight Controller | Personal Project

2026 – Present

- Designing a custom flight-controller PCB solo, applying FSAE power and layout experience to a dense, high-current mixed-signal board for a 5-inch FPV quadcopter built and tuned from individual components.

Technical Skills

Hardware Design: Component selection, schematic capture, PCB layout, DFM, assembly, bench bring-up, DC-DC converters, high-voltage isolation, analog signal conditioning

Tools: Altium Designer / Altium 365, LTspice, JLCPCB fabrication & PCBA, Git

Lab: SMD and through-hole soldering/rework, harness crimping & termination, oscilloscope, multimeter

Programming & Protocols: C++, Python; UART, SPI, I²C, ESP-NOW